Jordan University of Science and Technology

Faculty of Science

Department of Applied Chemistry

Experiment (1)

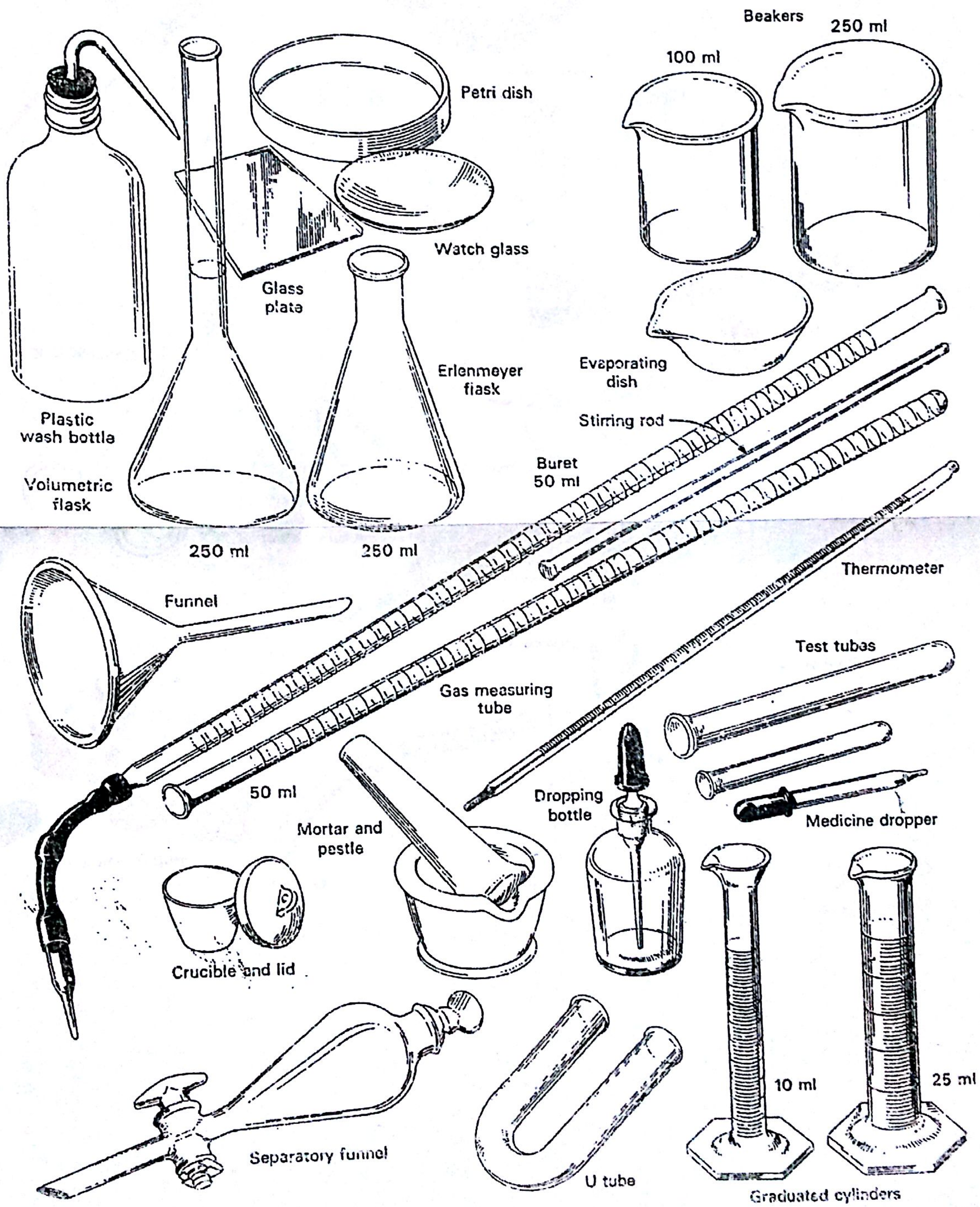
Density and Laboratory Tech.

Student Name	: هاشم احمد نجيب العقلة
I.D #	: 20152025009
Date of Experiment	:
Lab. Section	: 14
Partner Name	: مهند قاسم محمد خصاونه طارق محمد أديب بني يونس
Teacher Name	: آية الخطيب

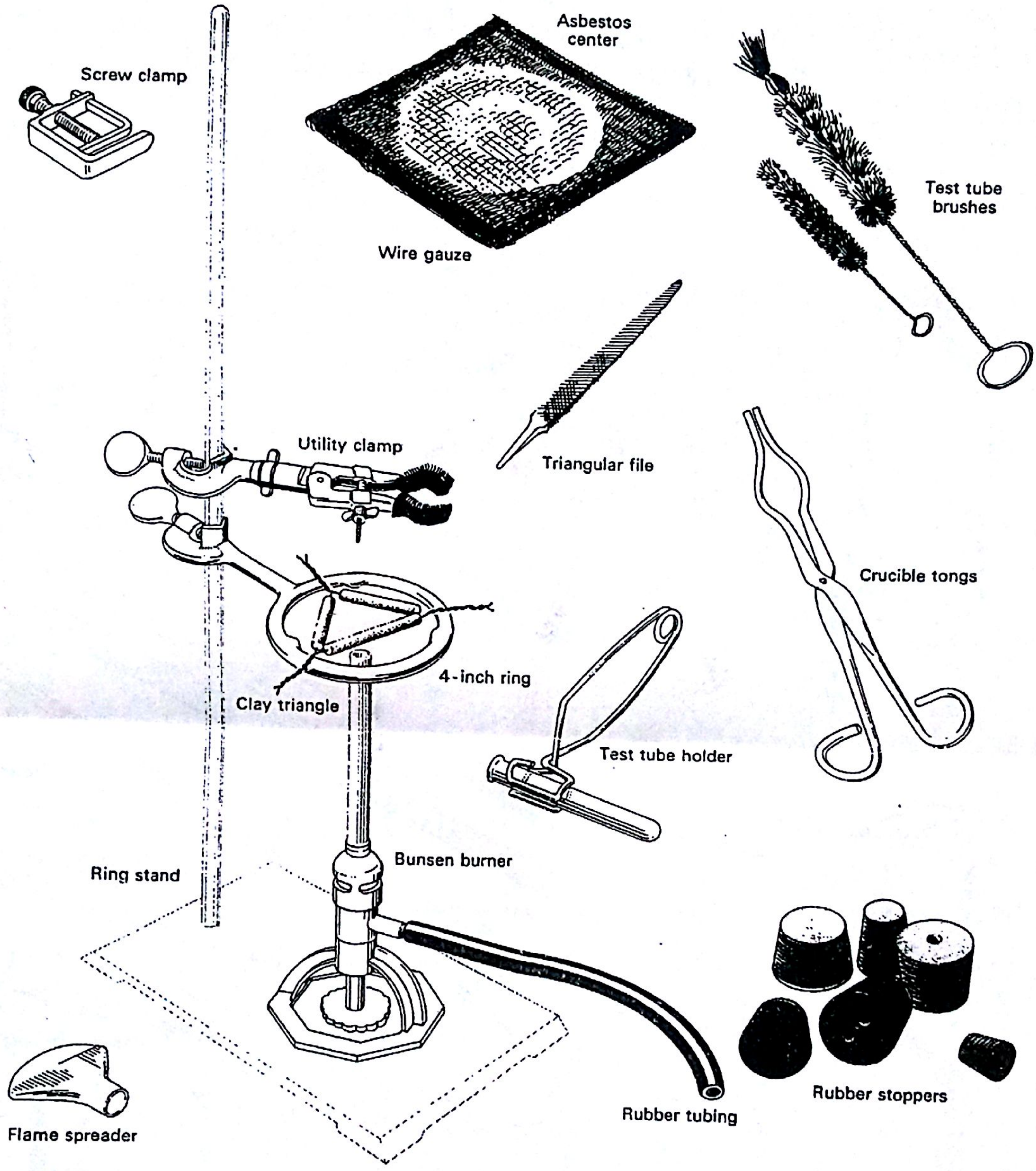
10/10

CHEM – LAB – 107

D. Names of Some Common Laboratory Equipment



Date


Screw clamp
Asbestos center
Wire gauze
Test tube brushes
Triangular file
Utility clamp
Crucible tongs
4-inch ring
Clay triangle
Test tube holder
Bunsen burner
Ring stand
Rubber stoppers
Rubber tubing
Flame spreader

Data and Results:

1- Unknown Irregular Solid

	Unknown Number: Trail (1)	Trail (2)
a- Mass of Solid	2.54 g	....
	....	....
b- Precision	± 0.01	±
c- Volume of water without solid	4.10 mL	....
	....	....
d- Precision	± 0.05	±
e- Volume of water with solid	4.40 mL	....
	....	....
f- Precision	± 0.05	±
g- Volume of solid	0.30 mL	....
h- Density of solid	8.5 g/mL	....
i- Average Density	....	....

1- Unknown Liquid

	Unknown Number: Trail (1)	Trail (2)
a- Mass of flask + stopper	20.78 g	....
b- Precision	± 0.01	±
c- Mass of flask + stopper + liquid	22.69 g	....
d- Precision	± 0.01	±
e- Mass of liquid	1.91 g	....
f- Volume of liquid	2.00 mL	....
g- Precision	0.05	....
h- Density of liquid	0.955 g/mL	....
i- Average Density	....	....

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Experiment (2)

Physical Separation of Mixtures

Student Name	: هاشم احمد نجيب العقله	
I.D #	: 20162025009	
Date of Experiment	:	
Lab. Section	: ~~8~~ 14	9.5
Partner Name	: مهند قاسم محمد خصاونه طارق محمد بني يونس عمر محمود حماوي	
Teacher Name	: آية الخطيب	

CHEM – LAB – 107

Data and Results:

Unknown No. P-7

A- Data

1- *Mass of evaporating dish.* 47.77 g

2- *Mass of evaporating dish and unknown mixture* 49.77 g

3- *Mass of evaporating dish & unknown mixture after heating* 48.98 g

4- *Mass of filter papers* 0.31 g

5- *Mass of filter papers & the solid* 0.47 g

B- Calculations and Results:

1- *Mass of original mixture* 2.00 g

2- *Mass of ammonium chloride in mixture* 0.79 g

3- *Mass of sodium chloride in mixture* 1.05 g

4- *Mass of silicon dioxide in mixture* 0.16 g

5- *% NH_4Cl in mixture* 39.5 %

6- *% NaCl in mixture* 52.5 %

7- *% SiO_2 in mixture* 8.0 %

* *Show your calculations.*

1 - total mass = 49.77 - 47.77 = 2.00 g

2 - mass of NH_4CL = 49.77 - 48.98 = 0.79 g

4 - mass of SiO_2 = 0.47 - 0.31 = 0.16 g

3 - mass of NaCL = 2.00 - (0.16 + 0.79) = 1.05 g

5 - % NH_4CL = $\frac{0.79}{2.00}$ * 100% = 39.5%

6 - % NaCL = $\frac{1.05}{2.00}$ * 100% = 52.5 %

7 - % SiO_2 (5) = $\frac{0.16}{2.00}$ * 100 % = 8%

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Faculty of Science

Department of Applied Chemistry

Experiment (3)

Limiting Reactant

Student Name	: هاشم احمد شبيب الحقله	
I.D #	: 20152025009	
Date of Experiment	: 7/3/2016	
Lab. Section	: 14	9 +
Partner Name	: [illegible] / محمد مومني / صهيب العمري	
Teacher Name	: آية الخطيب	

CHEM – LAB – 107

1. $n_{Ba_3(PO_4)_2} = \frac{m}{MW} = \underline{1.26 \times 10^{-3} \text{ moles}}$

2. 2 : 3 $\longrightarrow$ 1

$$\begin{array}{ccc} 3 & \longrightarrow & 1 \\ x & \longrightarrow & 1.26 \times 10^{-3} \end{array}$$

$x = \underline{3.79 \times 10^{-3}}$ moles

3. mass = MW * n = 0.925 g

$244.2 * 3.79 \times 10^{-3} = 0.925 \text{ g}$

4.

$$\begin{array}{ccc} 2 & \longrightarrow & 1 \\ x & \longrightarrow & 1.26 \times 10^{-3} \end{array}$$

$x = 2.52 * 10^{-3}$ moles

5. mass = MW * n = $2.52 \times 10^{-3} \times 380.2 =$ 0.958 g

6. 1 g

total mass   mass of ~~[illegible]~~ L-R

7. ~~1.0 − 0.925 = 0.075 g~~

$1.0 - 0.958 =$ ~~[illegible]~~

0.042 g

~~8~~

8. $\%_{BaCl_2 \cdot 2H_2O} = \frac{m}{\text{total mass}} \times 100\% = \frac{0.042}{1} = 4.2\%$

9. $\%_{Na_3PO_4 \cdot 12H_2O} = \frac{m}{\text{total mass}} \times 100\% = \frac{0.958}{1} \times 100\%$ = ~~95.8%~~ 95.8%

Data and Results:

	Trail 1	Trail 2
A- Precipitation of $Ba_3(PO_4)_2$:		
Unknown number	L-18	
1- Mass of salt mixture (g)	1.0g	
2- Mass of filter paper (g)	0.30g	
3- Mass of filter paper and $Ba_3(PO_4)_2$	1.06g	
4- Mass of $Ba_3(PO_4)_2$ (g) (1.06-0.3)	0.76g	
B- Determination of Limiting Reactant:		
1- Limiting reactant in salt mixture	PO_4^{-3}	
2- Excess reactant in salt mixture	Ba^{+2}	

Calculation:

1- Moles $Ba_3(PO_4)_2$ Precipitated: M.Wt. = 602.2 g/mol	1.26×10^{-3} mols	
2- Moles $BaCl_2.2H_2O$ reacted	3.79×10^{-3} mols	
3- Mass $BaCl_2.2H_2O$ reacted (g): M.Wt. = 244.2 g/mol	0.925 g	
4- Moles $Na_3PO_4.12H_2O$ reacted (g): M.Wt. = 380.2 g/mol	2.52 mols	
5- Mass $Na_3PO_4.12H_2O$ reacted (g): M.Wt. = 380.2 g/mol	0.958 g	
6- Mass of salt mixture A.1 (g)	1.0g	
7- Mass of excess reactant, B.2 (g)#6-#3or#5	0.042g	
8- Percent $BaCl_2.2H_2O$ in mixture	4.2%	
9- Percent $Na_3PO_4.12H_2O$ in mixture	95.8%	

* *Show your calculations for one trial on a separate sheet.*

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Department of Applied Chemistry

Experiment (4)

Chemicals In Everyday Life

Student Name	: هاشم احمد نجيب العكله
I.D #	: 2015 2025 009
Date of Experiment	:
Lab. Section	: 14
Partner Name	: مهند خصاونة ابراهيم عودة
Teacher Name	: ايه الخطيب

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CHEM – LAB – 107

Observations:

A- Glass Cleaner:

1- *Change of litmus paper* : From red to blue

2- *Reason for the change* : its a base (NH_3)

B- Fertilizer:

1- *Change of litmus paper* : From red to blue

2- *Describe the smell of gas* : pungent smell

3- *The smell is due to the formation of* : NH_3

C- Table Salt:

1- *Describe the smell of the gas formed* ...... pungent/strong ...

2- *The gas formed is* ...... HCL ...... *and it will turn a* ..... Blue ...... *litmus paper into* .... red ...

3- *When $AgNO_3$ solution is added to the salt solution, a* ... ~~AgCl~~ white ...... *solid is formed which is* ...... AgCL ..

D- Baking Soda:

1- *Observation from the addition of lemon juice or vinegar to baking soda:* ... escaping (evolve) of CO_2

2- *Observation and equation of holding a drop of $Ba(OH)_2$:* $CO_2 + Ba(OH)_2 \rightarrow BaCO_3 + H_2O$ (Forming of precipitate white $BaCO_3$)

E- Bathroom Cleaner:

1- *Observation and equation of addition of the cleaner to limestone:* evolve of CO_2

$HCl + CaCO_3 \rightarrow CO_2 + CaCl_2 + H_2O$

F- Bleach:

1- *Observation from the addition of bleach to KI solution and starch:* Forming a dark blue complex.

The equation is $Cl_2 + 2I \rightarrow 2Cl^- + I_2$ // ~~...~~ I_2 + Starch → dark

2- *The reaction is an oxidation-reduction, the element oxidized is:* ...... I^- ....

G- Drain Opener:

1- *Observation from the addition to warm water to the drain opener:* evolve of H_2

2- *The changes of litmus paper:* ..... from red to blue ....

H- Unknown: C-62

1- *Unknown No. is:* .... C-62 ....

2- *Unknown Ion:* ..... CO_3^{2-} ....

3- *Confirmatory tests:* (Colorless odorless gas CO_2 evolved.)

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Faculty of Science

Department of Applied Chemistry

Experiment (5)

Collegative Properties

Student Name	: هاشم احمد نجيب العقلة	
I.D #	: 20152025009	
Date of Experiment	: 21/3/2016	
Lab. Section	: 14	9
Partner Name	: مهند ضماوية ابراهيم عوده	
Teacher Name	: آية الخطيب	

CHEM – LAB – 107

$$\Delta T_f = K_f * m$$

$$1 = 1.86 * m$$

$$m = 0.538\ m$$

$$\text{molality} = \frac{\text{mol. solute}}{\text{kg solvent}}$$

Water

$1\ g = 1\ mL$

$$0.533 = \frac{n}{0.031}$$

$$n = 0.01667\ \text{mol}$$

$$n = \frac{\text{mass}}{\text{molar mass}}$$

$$mm = \frac{\text{mass}}{n}$$

$$= \frac{2.53}{0.01667} = 151.77\ g/mol$$

Data and Results:

	Trail 1	Trial 2
Part A. Freezing point of pure water:		
Volume of water	30 mL	
Freezing point of water	0 C°	

Part B. Freezing point of Solution;	Unknown	No: G-5
Volume of water	30.86 mL	
Mass of solute	2.53 g	
Freezing point of solutin	-1C°	

Calculations:

Molal freezing point depression constant for water = 1.86°C/m

mol

Density of water = 1.00g/ mL⁻¹

Mass of water in solution	≃ 0,0.31 kg
Mass of solute in solution	2.53 g
Molar mass of unknown	151.77 g/mol

** Show all your calculations for Trail 1.*

Data

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Department of Applied Chemistry

Experiment (6)

Acid – Base Titration

Student Name : هاشم احمد نجيب العقله

I.D # : 20152025009

Date of Experiment : ....

Lab. Section : 14

Partner Name : مهند قضاونه
ليث بدارنه

Teacher Name : آية الخطيب

9.5

CHEM – LAB – 107

Data and Results:

A- Standardization of NaOH solution:

	Trail 1	Trial 2
Mass of KHP (g)	0.43 g	....
Moles of KHP $n = \frac{m}{mm}$ (molar mass of KHP = 204.23g.mol)	0.00211 moles	....
Initial reading of NaOH buret (mL)	0 mL	....
Final reading of NaOH buret (mL)	13.8 mL	....
Volume of NaOH used (mL)	13.8 mL	....
Moles of NaOH used *	0.00211 moles	....
Molarity of NaOH (mol/L)	0.153 M	....
Average molarity of NaOH (mol/L)	....	....

B- Titrating an unknown solution of sulfuric acid **Unknown** **No: A - 35**

	Trail 1	Trial 2
Volume of H_2SO_4 solution used (mL)	20.0 mL	....
Initial reading of NaOH buret (mL)	13.8 mL	....
Volume of NaOH used (mL)	10.3 mL	....
Average molarity of NaOH (mol/L)	~~0.00~~	....
Moles of NaOH used	0.0016 moles	....
Moles of H_2SO_4	0.008 moles	....
Molarity of H_2SO_4 (mol/L)	0.04 M	....
Average molarity of H_2SO_4 (mol.L)	....	....

** Show your calculation for trial 1 in part A and part B.*

$M = \frac{n}{V}$

$n = \frac{n}{mm}$

* $M * V = \frac{m}{mm}$

$M * V = \frac{m}{mm}$

$M * 13.8 = 0.00211 \times 10^{-3}$

$M = 0.153$ M

$2 M_{H_2SO_4} * V_{H_2SO_4} = M_{NaOH} * V_{NaOH}$

$2 * 20 * M_{H_2SO_4} = 0.153 * 10.3$

0.0016

$M_{H_2SO_4} = 0.04$

* $Moles_{H_2SO_4} = 0.04 \times 20 \times 10^{-3}$

$= 0.008$ moles

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Faculty of Science

Department of Applied Chemistry

Experiment (7)

Determination of an Equilibrium Constant

Student Name : هاشم احمد نجيب العقلة

I.D # : 20152025009

Date of Experiment :

Lab. Section : 14

Partner Name :

Teacher Name : آية الخطيب

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CHEM – LAB – 107

20152025009

14 شعبة

1. St $[SCN^-]$ Calculations. هاشم العقله

$$[SCN]_{St} = \frac{M_1 V_1}{V_2} = \frac{0.002 * 1}{1+3+6}$$

trial#(2) حل تجربة *

$$= 2 * 10^{-4}\ M$$

$C_{st} = [FeSCN]_{st}$

$$\frac{A_{std}}{A_{eq}} = \frac{C_{std}}{C_{eq}}$$

2. $[Fe^{3-}]_{In}$: $\frac{M_1 V_1}{V_2} = \frac{0.005 * 1.5}{3+1.5+5.5} = 7.5 * 10^{-4}\ M$

3. $[SC\bar{N}]_{In}$: $\frac{M_1 V_1}{V_2} = \frac{0.002 * 3}{10} = 6.0 \times 10^{-4}\ M$

5. $[FeSCN^{-2}]_{eq} = C_{eq}$

$$\frac{A_{st}}{A_{eq}} = \frac{C_{st}}{C_{eq}} \quad \Big| \quad \frac{1.117}{0.378} = \frac{2*10^{-4}}{C_{eq}}$$

$$= 6.768 \times 10^{-5}\ M$$

6. $[Fe^{3+}]_{eq} = [Fe^{3+}]_{in} - [FeSCN^-]_{eq}$

$$7.5 * 10^{-4} - 6.768 \times 10^{-5}$$

$$= 6.8232 * 10^{-4}\ M$$

Excellent

7. $[SCN^-]_{eq} = [SCN]_{in} - [FeSCN^-]_{eq}$

$$6 \times 10^{-4} - 6.8232 * 10^{-4}\ M$$

$$= 5.3176 \times 10^{-4}\ M$$

8. $$K_{eq} = \frac{[FeSCN^{+2}]_{eq}}{[Fe^{+3}]_{eq}[SCN^-]_{eq}} = \frac{6.768 * 10^{-5}}{6.8232 * 10^{-4} * 5.3176 * 10^{-4}}$$

$$K_{eq} = 186.56496$$

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Report Sheet

Student Name ......هاشم الحطاب...... Student No. ..2015.2025009..

Instructor ......أ. و الخطيب...... Section No. ....14..............

Partner Name .... Date: ....

A. The Standard:

A=ab Conc.

	Trail 1	Trail 2	Trail 3
$[FeSCN^{2+}]_{std} = [SCN^-]_{in}$	2×10^{-4}		
Absorbance (A_{std})	1.117		

Average A_{std} =

B. The Equilibrium Mixture:

	Test Tube #1	Test Tube #2	Test Tube # 3	Test Tube # 4
$[Fe^{3+}]in$	1×10^{-3} M	7.5×10^{-4} M	5×10^{-4}	2.5×10^{-4}
$[SCN^-]in$	6×10^{-4} M	6×10^{-4} M	6×10^{-4} M	6×10^{-4} M
Absorbance, A	0.473	0.378	0.285	0.208
$[FeSCN^{2+}]_{eq}$	8.469×10^{-5} M	6.758×10^{-5} M	5.1029×10^{-5} M	3.7242×10^{-5} M
$[Fe^{3+}]_{eq}$	9.1531×10^{-4} M	6.8232×10^{-4} M	4.489×10^{-4} M	2.127×10^{-4} M
$[SCN^-]_{eq}$	5.1531×10^{-4}	5.3176×10^{-4} M	5.489×10^{-4} M	5.62758×10^{-4} M
K_{eq}	179.554	186.565	207.097	311.365

(Attached a separate paper that shows one sample calculations of K for solution # 2)

Average, Keq: (K1 + K2 + K3 + K4)/4

Standard Deviation, $\sigma_{n-1}, (Error) = \{[\sum(K_{eq} - K)^2]/(n-1)\}^{1/2} =$

$K_{eq} \pm \sigma_{n-1} =$

Rer

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Department of Applied Chemistry

Experiment (9)

Quantitative Yield of a Redox Reaction

Student Name : هاشم احمد نجيب الحقله

I.D # : 20152025009

Date of Experiment :

Lab. Section : 14

Partner Name : محمد المومني / عماد باجان / حسين العمري

Teacher Name : آية الخطيب

9.5

CHEM – LAB – 107

جمعية المهندسين الميكانيكيين
IME
Creativity is our brand
شعارنا الإبداع

Data and Results:

A: Data *Unknown NO:* R-87

Volume of $CuSO_4$ solution	10 *ml*
Mass of $CuSO_4$	10.4g *g*
Room temperature	25 *C°*
Mass of filter paper	0.28 *g*
Mass of Filter paper + Cu	0.51 *g*
Mass of Cu	0.23 *g*

Cu
63.55

B- Results (show your calculation methods, with units):

Density of $CuSO_4$ solution	1.04 g/mL
Molarity of $CuSO_4$ solution	0.362 M
% by mass of Cu in solution	2.21 %

* *Write the net ionic equation for the replacement of Cu^{2+} by Zn^{2+} in aqueous solution:*

$Cu^{+2}_{aq} + SO^{-2}_{4\,aq} + Zn_{(s)} \rightarrow Cu_{brown} + Zn^{+2}SO^{-2}_{4\,aq}$

Blue

* *Write the net ionic equation for the dissolution of zinc by dilute HCl:*

$Zn_{(s)} + HCl_{(aq)} \longrightarrow ZnCl_2 + H_{2(g)}$

mass of Cu = 0.51 − 0.28 = 0.23

mass of $CuSO_4$ ⇒ $d = \frac{m}{V}$

$m = d * V = 1.04 * 10 = 10.4g$

$Molarity = \frac{n}{V}$

$n \Rightarrow \frac{m}{mm} = \frac{0.23}{63.55} = 3.62 \times 10^{-3}$ moles

$\% \text{ mass} = \frac{\text{Cu mass}}{\text{Sd. mass}} \times 100\% = \frac{0.23}{10.4} \times 100\% = 2.21\%$

$Molarity = \frac{3.62 \times 10^{-3}}{10 \times 10^{-3}} = 0.362\,M$

Exp: 11 Rate Law. Partner: ليث بدارنة

Report Sheet:

Sec: 14

Name: هاشم احمد نجيب العملة *No.:* 2015202009 *Date:*

A. Data		١	٨	٩	
Solution Temp.	1	2	3	4	5
1. Solution Temperature (°C)		25°C	25°C	25°C	
2. Time elapsed for color change, t (seconds)		75s	44s	21 s	

10

B. Calculations					
1. Mol $S_2O_3^{2-}$ reacted		10^{-4} Mol	10^{-4} Mol	10^{-4} Mol	
2. Mol I_2 produced		5×10^{-5} Mol	5×10^{-5} Mol	5×10^{-5} Mol	
3. rate = mol I_2/time(s) (molr/sec)		6.6×10^{-7}	1.136×10^{-6}	2.38×10^{-6}	(M/s)
4. $[I^-]_i$ (mol/L)		0.03M	0.045 M	0.045 M	
5. $[H_2O_2]_i$ (mol/L)		0.015M	0.015 m	0.025 M	

C. Determination of reaction order

Slope = p = 1

Slope = q = 1

Rate law of the reaction: rate $= k'[I^-]^1[H_2O_2]^1$

D. Determination of k'

	1	2	3	4	5
1. Values of k'		1.5×10^{-3}	1.7×10^{-3}	$\cancel{\ }\times10^{-3}$	

$M^{-1}S^{-1}$

2. Average value of k' 1.77×10^{-3} $M^{-1}S^{-1}$ 2-1

E. Show your calculations for solution 1

هاشم احمد نجيب العقله

$Mol\ S_2O_3^{-2} = MV$

$= 0.02 \times 5\times10^{-3} = 1\times10^{-4}$ mol

$Mol\ I_2 = \frac{1}{2}\ Mol\ S_2O_3$

$= 5\times10^{-5}$ mol

trial 2

$[I^-] = \frac{MV}{V_{all}} = \frac{0.3\times10}{100} = 0.03\,M$

$rate = mol\ I_2/\Delta t$

$= 5\times10^{-5}/75 = 6.6\times10^{-7}\ M/s$

[redacted]

$[H_2O_2]\ \frac{0.1M * 15}{100} = 0.015\,M$

$p \rightarrow \quad \frac{R_3}{R_2} = \frac{[I^-]^p_3}{[I^-]^p_2}$

$\frac{11.36 \times 10^{-7}}{6.6 \times 10^{-7}} = \left(\frac{0.045}{0.03}\right)^p$

$1.7 = (1.5)^p$

$\log 1.7 = p \log 1.5 \qquad \underline{\underline{p \cong 1}}$

$q \rightarrow \quad \frac{R_3}{R_4} = \frac{[H_2O_2]^q_3}{[H_2O_2]^q_4}$

$\frac{10^{-6} \times 1.136}{10^{-6} \times 2.38} = \left(\frac{0.015}{0.025}\right)^q$

$(0.47) = (0.6)^q$

$\log 0.47 = q \log 0.6$

$q = \frac{\log 0.47}{\log 0.6}$

$\underline{\underline{q}} \cong 1.47 \cong 1 \qquad q \cong 1$

$k' \Rightarrow \quad R = k'[I^-][H_2O_2]$

② $k'_2 \Rightarrow 1.5 \times 10^{-3} \ M^{-1}s^{-1}$

$k_3 \Rightarrow 1.7 \times 10^{-3} \ M^{-1}s^{-1}$

$k_4 \Rightarrow 2.1 \times 10^{-3} \ M^{-1}s^{-1}$

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Faculty of Science

Department of Applied Chemistry

Experiment (10)

Qualitative Analysis of Cations

Student Name	: هاشم احمد نجيب القلعه
I.D #	: 2015202500 9
Date of Experiment	:
Lab. Section	: 14
Partner Name	صهيب الحمري / ليث بدارنة / عمر حمادي
Teacher Name	: آية الخطيب

10

CHEM – LAB – 107

Unknown Number: CA-03

Mark each action below to indicate if it is present (P) or absent (A) in your unknown.

a- Mg^{2+} …………………………	*d-* Pb^{2+} Present
b- Ba^{2+} …………………………	*e-* Hg_2^{2+} Absent
c- Ca^{2+} …………………………	*f-* Ag^{+} present